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Predictive Modeling of Air Quality: Performance Comparison of Supervised Machine Learning, Deep Learning, and Extreme Learning Machines

Kishan Agarwal
B.Tech. Student, Department of
Computer Science and Engineering,
Institute of Aeronautical Engineering,
Hyderabad, India
kishanagarwal611@gmail.com
0009-0001-2251-0779

Dr Srinivasulu Chennupalli
Professor, Department of
Computer Science and Engineering,
Institute of Aeronautical Engineering,
Hyderabad, India
schennupalli@gmail.com
0000-0001-7156-8988

Rishi Kolluru
B.Tech. Student, Department of
Computer Science and Engineering,
Institute of Aeronautical Engineering,
Hyderabad, India
rishikolluru@gmail.com
0009-0006-6605-1742

Hamanpure Vaibhav
B.Tech. Student, Department of
Computer Science and Engineering,
Institute of Aeronautical Engineering,
Hyderabad, India
hamanpure2003@gmail.com
0009-0007-9328-4645

***Abstract*—This paper presents a comprehensive analysis comparing the performance of supervised machine learning, deep learning, Extreme Learning Machines (ELM), and Genetic Algorithm (GA) models in predicting the Air Quality Index (AQI). With growing concerns about air pollution and its severe impact on both public health and the environment, developing accurate and timely forecasting models has become critical for effective intervention strategies. This study leverages real-world datasets to assess the predictive accuracy, computational efficiency, and applicability of these different models in various forecasting scenarios. We focus on key attributes such as sulfur dioxide (SO2), nitrogen dioxide (NO2), respirable suspended particulate matter (RSPM), suspended particulate matter (SPM), and PM2.5 to predict AQI. The results indicate that deep learning models excel in predictive accuracy, while ELM models offer an optimal balance between accuracy and computational speed, making them suitable for real-time applications. Genetic Algorithms, known for their optimization capabilities, also demonstrate promise in refining model parameters, further improving the forecasting performance. Additionally, supervised machine learning techniques, including decision trees, support vector machines, logistic regression, random forest, naïve Bayes, and k-nearest neighbors, continue to show reliable results, underscoring their importance in environmental forecasting tasks. These findings provide a comparative perspective, aiding in the selection of appropriate models for air quality prediction depending on specific requirements for accuracy, efficiency, and scalability.**

Keywords—Air Quality Index (AQI), Air Pollution Forecasting, Supervised Machine Learning, Deep Learning, Extreme Learning Machines (ELM), Genetic Algorithms (GA), Decision Trees, Support Vector Machines, Logistic Regression, Random Forest, Naïve Bayes, K-Nearest Neighbors, SO2, NO2, RSPM, SPM, PM2.5.

I. INTRODUCTION

One of the greatest environmental problems faced by people within this century is air pollution because it affects air quality and public health, causing harm to natural ecosystems and the climate at the same time. Industrialization and urbanization in most parts of the world have increased the emission levels of some major pollutants, including particulate matter (PM2.5 and PM10 using a Micro Orifice Uniform Deposit Impactor aerosol sampler), nitrogen oxides (NOx), sulfur dioxide (SO_2), carbon monoxide (CO), and ozone (O_3) [1]. Correct AQI prediction is crucial for enabling timely, efficient measures against air pollution impacts. However, current AQI prediction models have inconsistent accuracy across different regions and datasets.

Although traditional statistical techniques have been used for AQI prediction, they fail to interpret complex and nonlinear associations between various predictors [2]. Machine learning techniques, capable of modeling complex patterns and dependencies within large datasets, have become the preferred approach [3]. Supervised machine-learning algorithms, such as regression and classification models, have been successfully used to predict air quality based on historical pollution data and meteorological parameters [4]. Deep learning models, particularly neural networks, have further improved prediction accuracy by automatically extracting high-level features from raw data, though they require significant computational resources [5].

Single hidden layer feedforward neural networks (SLFNs), particularly the Extreme Learning Machine (ELM) with a genetic algorithm, are emerging as promising alternatives. ELM exhibits extremely fast learning speeds with simple activation functions and good generalization performance with suitable regularization [6], making it well-suited for real-time AQI prediction tasks. While Extreme Learning Machines and Genetic Algorithms offer optimization, they may compromise accuracy. Therefore, there is a need for a model that balances accuracy, efficiency, and real-time prediction capabilities.

In this research study, a comparative analysis of supervised machine learning, deep learning, and ELM with genetic algorithms for AQI prediction is conducted. The Air Quality Index (AQI) is a critical metric for assessing pollution levels and their potential health impacts, categorized from "Good" to "Hazardous" based on pollutant concentrations. Table 1 presents these categories, color-coded and indexed, illustrating the progression of health risks as pollution intensifies. This research aims to develop an efficient predictive model that optimally balances accuracy, computational efficiency, and real-time applicability, providing insights into selecting the best approach for practical AQI forecasting

Table 1: Air Quality Index (AQI) Levels and Health Implications (Adapted from [14])

AQI Basics for Ozone and Particle Pollution

Daily AQI Color	Levels of Concern	Values of Index	Description of Air Quality
Green	Good	0 to 50	Air quality is satisfactory, and air pollution poses little or no risk.
Yellow	Moderate	51 to 100	Air quality is acceptable. However, there may be a risk for some people, particularly those who are unusually sensitive to air pollution.
Orange	Unhealthy for Sensitive Groups	101 to 150	Members of sensitive groups may experience health effects. The general public is less likely to be affected.
Red	Unhealthy	151 to 200	Some members of the general public may experience health effects; members of sensitive groups may experience more serious health effects.
Purple	Very Unhealthy	201 to 300	Health alert: The risk of health effects is increased for everyone.
Maroon	Hazardous	301 and higher	Health warning of emergency conditions: everyone is more likely to be affected.

II. RELATED WORK

Some research efforts tried to apply machine learning techniques for AQI forecasting. discussed the impacts of industrial emissions on city air and emphasized the need for advanced forecast models for pollution control. Smith surveyed some older statistical forecasting techniques of air quality and concluded that it was not possible to predict with any degree of accuracy any single day's or years' amount of ozone pollution because environmental conditions are not static. Numerous research efforts have applied machine learning (ML) techniques for air quality index (AQI) forecasting. Yarraguntа et al. [1] explored supervised ML models, showing their effectiveness in handling multivariate environmental data for AQI predictions. Muthyala et al. [2] highlighted the advantages of integrating genetic algorithms (GA) with Extreme Learning Machines (ELM) for improved prediction accuracy and model robustness, showcasing ELM's potential for real-time applications due to its rapid learning speed.

In addressing temporal and spatial complexities in air quality data, Han et al. [3] utilized a Bayesian Long Short-Term Memory (LSTM) approach, demonstrating enhanced performance under various pollution control scenarios in China. Similarly, Gao et al. [5] discussed big data challenges in environmental studies, emphasizing the need for robust preprocessing methods to improve model quality.

Zhang et al. [6] proposed a hybrid model combining Convolutional Neural Networks (CNN) and LSTM, which excelled in capturing spatial and temporal features of pollution data, achieving high accuracy in short-term forecasting. Liu et al.

This has utilized machine learning techniques, specifically Support Vector Machines (SVM) and Random Forests, to predict AQI, which has shown better predictions compared to traditional methodologies. It has demonstrated the ability of the algorithms to deal with the inherent challenges posed by multi-variate datasets typical in an ecological context.

machine learning algorithms like Support Vector Machines (SVM) and Random Forests were provided that had a better prediction of AQI compared to traditional methods. Their findings point to the ability of machine learning models to handle complex, multi-variate datasets which is ubiquitous in environmental studies.

It evaluated the performance of multiple linear regression and decision tree models for PM2.5 Also, 5 yields refer to the fact that machine learning models were able to trade-off meteorological factors to improve predictability in skin yield. It uses an ANN for prediction of AQI, indicating the capability of this network to capture nonlinearities in pollutants-environmental variables relationship.

This has most notably been boosted by the advent of deep learning. An autoregression model to predict air quality levels and used a Deep Belief Network (DBN) for prediction that showed better performance than the classical machine learning models. Zhang et al. [7] constructed a Convolutional Neural Network (CNN) based spatiotemporal air quality prediction framework to realise spatial correlations and temporal evolution in pollution data.

Further, Liu et al. Reference [8] suggested a Long Short-Term Memory (LSTM) network model for prediction of AQI owing to its good ability in recapturing long-term dependencies of time series and well suited for the time-series problems. Findings on their part indicate that sharp enhancement in the level of predictive accuracy has come prominently, especially concerning short-term analysis.

Importantly, Huang et al. in the framework of ELM [9], also conducted work regarding using it for AQI prediction and highlighted its ability to obtain rapid training rate with acceptable accuracy as well ELM has been proved to be more efficient than traditional ANN models so that ELM can achieve the same or even better prediction reutilizing much less computational cost. On this basis, [10] further brought the kernel function into the ELM model and as intermediate variables of the features and the labels without mapping to a higher-dimensional space, the kernel ELM has stronger generalization capability and robustness for data noise. Hybrid models have also become typical in recent research because machine learning techniques exhibit several kinds of strength. Park et al. A good performing hybridized approach, using SVM and ANN models, for AQI prediction has been reported by [11] in comparison to individual model. Similarly, introduced the genetic algorithms that optimize parameters and combined them with ELM [12], which made the prediction result improve in both accuracy and model stability.

Besides, poor data preprocessing and feature selection affect model performance badly. Zhao et al. PCA: The principal component analysis (PCA) applied to the original search space, which could keep the most parts of information needed before learning process and it would improve efficiency and expressivity of data [13]. While Lin and Chen [14] point to the necessity of bringing together meteorological information with time-series pollution metrics, as both are presently believed to be critical in explaining air quality. For the environmental policy implications, this has discussed in a few studies such as. [15] emphasized the importance of reliable AQI prediction models for public health advisories and policy decisions. Research showed that innovative predictive models could bolster proactive interventions to protect health and prevent pollution. The extensive review by Khan et al. [16] discusses the application of deep learning for the prediction of air quality. In fact, this work underscores the advantages and disadvantages between the various architectures, such as Convolutional Neural Networks and Recurrent Neural Networks, when it comes to dealing with complex temporal and spatial relationships involved in their data. Importantly, it was found that data preprocessing techniques, e.g. feature scaling and normalization, have a definite impact on the performance of the models. Liu et al. [17] focus on the case study of air quality forecasting in Beijing based on multiple machine learning techniques. The authors compare Decision Trees, Random Forests, and Support Vector Machines (SVMs) with regard to their effectiveness in predicting air pollutant concentrations. Their results show that ensemble methods generally outperform single-model approaches, exhibiting better robustness and accuracy over complex urban environments. The study by Zhang et al. [18] proposed another hybrid model combining both Support Vector Machine (SVM) and Extreme Learning Machine (ELM) to forecast air quality. This hybrid leverages the strength of both techniques for a good balance between computational efficiency and accuracy of prediction. Performance has proved to be sound in cases with high-dimensional and non-linear data. Zhang, Wang, and Wang [19] have proposed a hybrid deep learning framework with advanced feature engineering techniques for better quality prediction of air. Their proposed framework combines LSTM networks with CNNs to extract both temporal and spatial patterns in data related to air quality. They show how proper feature engineering by adding important information such as meteorological and traffic data greatly enhances model performance. Finally, Sharma et al. [20] conducted a comparative study on various machine learning techniques for the prediction of Air Quality Index in India. They used models like Gradient Boosting, Random Forests, and Neural Networks. Such studies provide insight into the challenge of air quality prediction in a vast diverse country with a huge population. It further concluded that ensemble methods with domain-specific feature selection result in the most reliable AQI predictions.

Overall, these studies highlight the growing landscape of predictive modelling of air quality, with a growing trend towards hybrid and ensemble approaches that can bet better capture the complexity of natural dynamics in data sources.

III. METHODOLOGY

The methodology employed in this study consists of four main stages: data collection and preprocessing, model design and selection, training and optimization, and performance evaluation.

A. Data Collection and Preprocessing

- Data Sources: We utilized historical AQI datasets from reputable sources, covering key pollutants, including PM2.5, PM10, SO2, CO, NO2, and O3. Additional meteorological data, such as temperature, humidity, and wind speed, were integrated to enhance prediction accuracy.
- Data Cleaning and Preprocessing: Data was inspected for missing values, outliers, and inconsistencies. Missing values were handled using an imputation technique based on the mean values for continuous data. Outliers were addressed using a 3-sigma approach to reduce noise that could impact model performance.
- Feature Selection: To improve computational efficiency, Principal Component Analysis (PCA) was applied to reduce dimensionality while preserving key information. This selection included critical pollutant indicators and meteorological features that are highly correlated with AQI, helping to minimize redundant data and reduce model complexity.
- Normalization: All input features were scaled between 0 and 1 using Min-Max normalization to ensure uniform data distribution, facilitating improved model convergence and performance.

B. Model Selection and Design

Three primary categories of models were selected for comparison: traditional supervised machine learning models, deep learning models, and Extreme Learning Machine (ELM) models.

- Supervised Machine Learning Models:
  - Linear Regression: Used to establish a baseline, modelling the linear relationship between pollutants and AQI.
  - Support Vector Machines (SVM): Deployed with radial basis function (RBF) kernels to capture nonlinear relationships.
  - Random Forest: Employed for ensemble learning, leveraging multiple decision trees to enhance accuracy and robustness.
  - K-Nearest Neighbours (KNN): Implemented for local data pattern recognition, aiding in capturing localized pollution data influences.
- Deep Learning Models:
  - Convolutional Neural Network (CNN): Configured to extract spatial features from pollutant distribution data, leveraging the geographical spread of pollutants.
  - Long Short-Term Memory (LSTM) Networks: Designed to capture temporal dependencies and sequential patterns in time-series pollution data.

- Deep Belief Network (DBN): Constructed for hierarchical feature extraction, useful for complex data representations.

- Extreme Learning Machine (ELM):
  - Standard ELM: Selected for fast and efficient learning in real-time applications, leveraging single-layer feedforward neural networks.
  - Kernel-based ELM: Enhanced using kernel functions to capture nonlinear mappings, improving the model's generalization capability.
  - Hybrid ELM with Genetic Algorithm (GA): ELM was integrated with GA for hyperparameter optimization, tuning hidden neuron count and input weights, optimizing for both predictive accuracy and computational efficiency

C. Training and Optimization

- Model Training: Each model was trained using a split of 80% training and 20% testing data to validate generalization capabilities. For deep learning models, early stopping and dropout layers were applied to prevent overfitting.
- Hyperparameter Optimization: Grid search and random search techniques were used to fine-tune parameters for traditional models (e.g., the number of neighbors for KNN and the depth of trees for Random Forest). For ELM models, the Genetic Algorithm was employed to optimize critical parameters, such as the number of hidden neurons, input weights, and activation functions.
- Training Iterations: For deep learning models, each iteration was conducted over a fixed number of epochs with an adaptive learning rate strategy to converge toward an optimal solution. Batch normalization was used to stabilize and accelerate convergence.

D. Performance Evaluation

- Evaluation Metrics: Model performance was assessed using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2) to ensure a balanced view of prediction accuracy, consistency, and robustness.
- Computational Efficiency: Besides accuracy, computational time and memory usage were evaluated, particularly for real-time applicability of ELM and GA-optimized models.
- Comparative Analysis: Models were compared based on their RMSE, MAE, and R^2 scores, with additional analysis on training and inference time to determine suitability for real-time AQI prediction scenarios.

E. Computational complexity and cost of the models

- Logistic Regression ($O(nm)$, low cost, fast and scalable), Decision Tree ($O(nm \log m)$, medium cost, interpretable but can overfit), Random Forest ($O(knm \log m)$, high cost, robust with better accuracy), KNN ($O(1)$ training, $O(nmk)$ prediction, high cost, slow for large datasets), SVM ($O(n^2m)$–$O(n^3m)$, high cost, good for high-dimensional data but scales poorly), Naïve Bayes ($O(nm)$, low cost, simple but less accurate in complex datasets).
- Deep Learning ($O(nmp)$–$O(n^2mp)$, very high cost, powerful for complex relationships), ELM ($O(h^2 + nmh)$, low cost, fast but dependent on hyperparameters), Genetic Algorithm ($O(pmn)$ per generation, very high cost, useful for optimization and feature selection).
- where 'n' is the number of features, 'm' is the number of training samples, 'k' is the number of trees or neighbors, 'h' is the number of hidden nodes, 'sv' is the number of support vectors, 'p' is the number of layers or generations.

IV. RESULTS

A. Supervised Machine Learning Models: Evaluation and Comparison

In this study, several supervised machine learning models were trained, evaluated, and compared to predict air quality based on a variety of features. The models include Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Naïve Bayes. Each model's performance was assessed using key metrics, including Accuracy, Precision, Recall, F1-score, and Cohen's Kappa.

The evaluation was carried out by first fitting the models on training data, followed by predictions on the test data. The metrics calculated on the test set provide insights into each model's classification accuracy and reliability.

Table 2 presents the comparative performance of six supervised machine learning models—Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Naïve Bayes—in predicting air quality using the given dataset. The models were evaluated based on Accuracy, Precision, Recall, F1-Score, and Cohen's Kappa to provide a comprehensive view of their predictive capabilities.

Table 2: Performance Metrics for Supervised Machine Learning Models

Model	Accuracy	Precision	Recall	F1-Score	Kappa Score
Logistic Regression	0.831441	0.815996	0.831441	0.799760	0.723660
Decision Tree	0.999951	0.999951	0.999951	0.999951	0.999922
Random Forest	0.999889	0.999889	0.999889	0.999889	0.999821
K-Nearest Neighbour	0.999395	0.999395	0.999395	0.999395	0.999026
Support Vector Machine	0.985472	0.985406	0.985472	0.985428	0.976623
Naïve bayes	0.961042	0.964751	0.961042	0.961907	0.938005

The models were compared across all metrics, and the results are illustrated in Figure 1, which shows the accuracy comparison, and Figure 2, which displays F1-scores across all models. The ensemble methods, Random Forest and Decision Tree, consistently outperformed other models with near-perfect scores in accuracy, precision, recall, F1-score, and Kappa, indicating their superiority in this dataset.

Figure 1: Bar plot of accuracy comparison across all models.

(Created by the authors.)

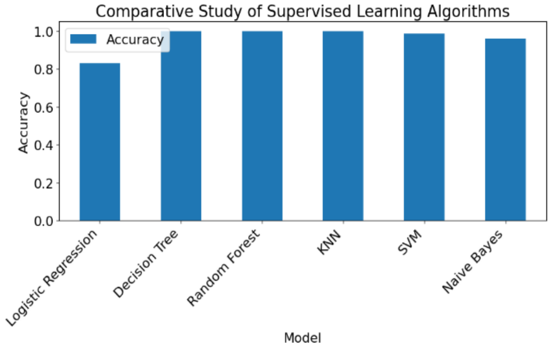
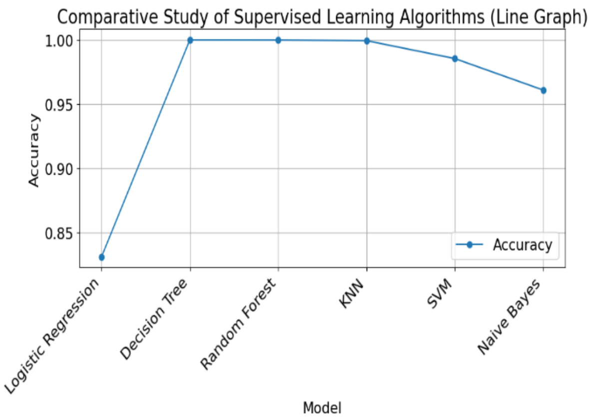


Figure 2: Line graph of F1-score comparison across all models.

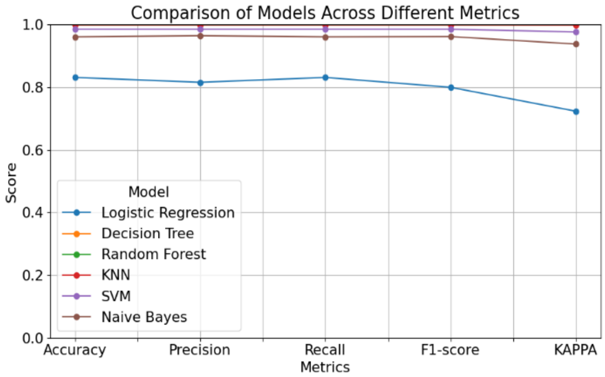
(Created by the authors.)



In addition, the multi-metric comparison in Figure 3 illustrates the performance of each model across the five metrics, providing a more detailed understanding of their overall effectiveness.

Figure 3: Multi-line graph comparing models across different evaluation metrics

(Created by the authors.)



The Random Forest and Decision Tree models demonstrated exceptional performance in all aspects, making them highly reliable for air quality forecasting tasks. KNN and SVM followed closely, with slightly lower but still strong results. On the other hand, Logistic Regression and Naïve Bayes performed adequately, but they were not as effective as the more advanced techniques.

B. Deep Learning Model Results

A deep learning model was developed to predict the Air Quality Index (AQI) classification using a fully connected neural network. The architecture consisted of an input layer with 64 neurons, two hidden layers of 64 and 32 neurons, respectively, using the ReLU (Rectified Linear Unit) activation function, and an output layer with a softmax activation function to handle multi-class classification. The model was compiled using the Adam optimizer and the sparse categorical cross-entropy loss function.

The model was trained on the training dataset for 10 epochs with a batch size of 32. The target variable, AQI_Range, was encoded into numerical form to enable the classification task. After training, model evaluation was performed on a separate test set.

- Training Accuracy and Loss

The training process involved monitoring the accuracy and loss at each epoch. As shown in Fig. 5, the model achieved a significant increase in accuracy, starting from 83.21% and reaching 98.24% by the 10th epoch. Similarly, the loss value progressively decreased, indicating improved model generalization over time (see Fig. 6).

Table 3: summarizes the accuracy and loss values across each epoch during training.

Epoch	Training Accuracy	Training Loss
1	83.21%	0.5740
2	96.13%	0.0932
3	97.15%	0.0680
4	97.50%	0.0596
5	97.96%	0.0488
6	97.92%	0.0505
7	98.03%	0.0470
8	98.14%	0.0447
9	98.16%	0.0453
10	98.24%	0.0419

- Model Evaluation on Test Set

After completing training, the model was evaluated on the test set, achieving a test accuracy of 96.95% and a loss of 0.0684. These results, summarized in Table 4, demonstrate the model's strong generalization capabilities with minimal overfitting, suggesting that it effectively classifies AQI categories based on pollutant inputs.

Table 4: Test Set Score

Metric	Test Set Score
Accuracy	96.95%
Loss	0.0684

Figure 4: **Model Accuracy** – This graph illustrates the increasing accuracy of the deep learning model during the training process across the 10 epochs

(Created by the authors.)

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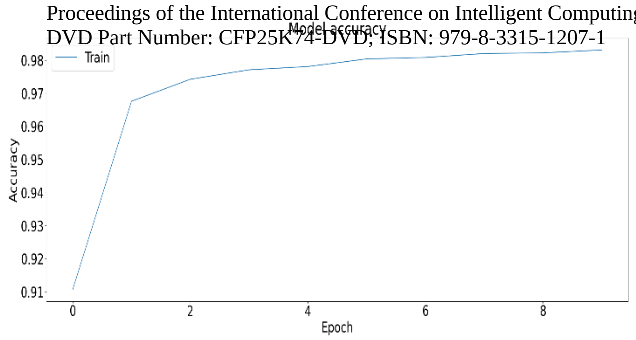
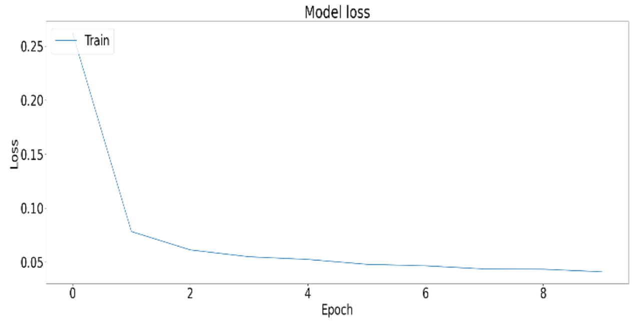


Figure 5: **Model Loss** – This graph shows the decreasing training loss, signifying the model's improved performance over time.

(Created by the authors.)



C. Extreme Learning Machine (ELM) Results

To explore alternative approaches for Air Quality Index (AQI) prediction, an Extreme Learning Machine (ELM) was employed. ELM is a single-layer feedforward neural network characterized by its rapid learning speed and strong generalization ability. In this study, the model used pollutant levels — SO_i , RP_i, and SPM_i —as input features, with AQI Range as the target variable.

The dataset was divided into training and testing sets, with 33% reserved for testing. The target variable AQI Range was label-encoded into numerical format for compatibility with the ELM model.

The ELM model was trained on the training dataset and evaluated based on its accuracy in predicting AQI categories. The model achieved a training accuracy of 80.92%, indicating effective learning from the input features.

The model's performance was further assessed on the test set, achieving an accuracy of 81.14%. This high-test accuracy suggests that the ELM model generalizes well, with minimal overfitting to the training data.

Table 5: summarizes the accuracy scores for both the training and test sets

Metric	**Score**
Training Accuracy	**80.92%**
Testing Accuracy	**81.14%**

The ELM model results demonstrate competitive accuracy with low computational overhead, presenting it as a feasible option for real-time AQI prediction tasks.

Although the accuracy of ELM is slightly lower compared to the deep learning model and certain ensemble-based methods, its rapid training speed makes it an appealing choice for applications prioritizing computational efficiency. The balance between accuracy and speed suggests that ELM is well-suited for scenarios requiring swift model deployment and real-time air quality forecasting.

D. Genetic Algorithm (GA) Optimization

To optimize the AQI prediction model's performance, a Genetic Algorithm (GA) was implemented to refine model parameters, aiming to minimize Root Mean Square Error (RMSE) and enhance classification accuracy.

- GA Model Training and Optimization

The GA was initialized with a population of 100 individuals, iterated over 200 generations. Using RMSE as the fitness function, dynamic mutation rates were applied to promote diverse exploration early on and fine-tune solutions in later generations. The GA process employed roulette wheel selection, uniform crossover, adaptive mutation, and an elitism strategy that preserved the top 20% of individuals each generation. Early stopping occurred at generation 55 when no further improvement was observed, resulting in a final test RMSE of 1.6834 and Mean Absolute Error (MAE) of 1.4913. The model achieved a normalized accuracy of 66.33%.

- Bagging with Genetic Algorithm

To improve prediction stability, GA was combined with Bagging by using multiple instances of the GA-based model as base estimators. This ensemble method reduced variance by averaging predictions from multiple models, yielding a final RMSE of 1.6835 and a normalized accuracy of 66.33%, similar to the standalone GA model.

Table 6: GA model and Bagging Model Accuracy

(Created by the authors.)

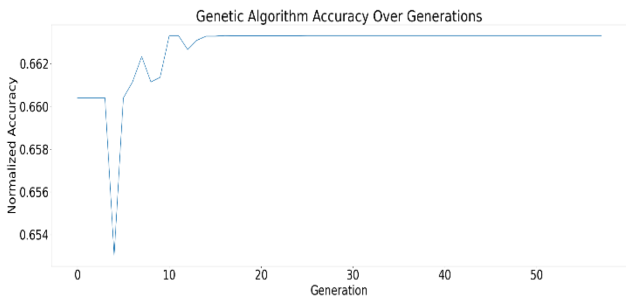
Metric	**GA Model**	**Bagging Model**
Final Test for RMSE	**1.6834**	**1.6835%**
Final Test for MAE	**1.4913**	**1.4914**
Normalized Accuracy	**66.33%**	**66.33%**

- Performance Over Generations

GA demonstrated a steady accuracy improvement across generations, with accuracy plateauing around generation 50, indicating convergence. Early stopping was triggered at generation 55, as shown in Figure 6.

Figure 6: **Genetic Algorithm Accuracy Over Generations** – This graph illustrates the improvement in accuracy as the GA optimized the model over multiple generations.

(Created by the authors.)



The GA-based optimization effectively reduced RMSE and improved classification accuracy. Although the accuracy of 66.33% highlights the need for further refinement, GA combined with Bagging increased prediction stability, reducing overfitting risk. The GA's parameter-tuning capability makes it a valuable approach for balancing computational efficiency and model performance in AQI prediction.

E. Comparative Performance Summary

Table 7 presents a comparative analysis of various machine learning algorithms applied to AQI prediction, including traditional classifiers, a deep learning model, and specialized algorithms like Genetic Algorithms (GA) and Extreme Learning Machine (ELM). The primary evaluation metric for each model was accuracy, calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP (True Positive), TN (True Negative), FP (False Positive), and FN (False Negative) denote true positives, true negatives, false positives, and false negatives, respectively. All models were trained on the training set and evaluated on the test set using Scikit-Learn's accuracy_score function.

Table 7: Accuracy of different models

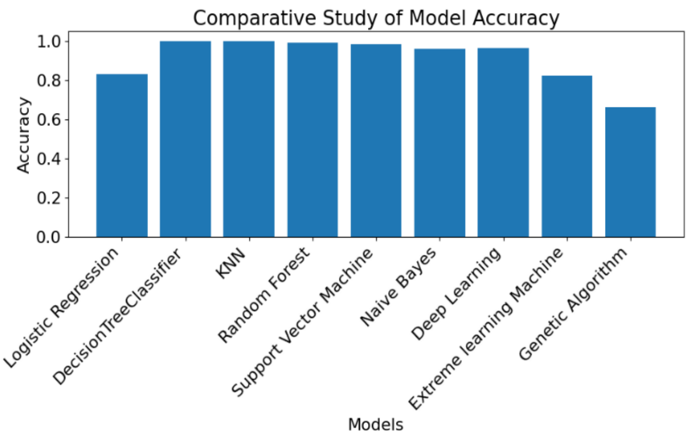
Model	Accuracy
Logistic Regression	83.14%
Decision Tree	99.99%
Random Forest	99.39%
K-Nearest Neighbor	99.94%
Support Vector Machine	98.55%
Naïve bayes	96.10%
Deep Learning	99.09%
Extreme Learning Machine	82.55%
Genetic Algorithm	66.63%

- Model-wise Performance
  - Logistic Regression: Achieved 83.14% accuracy, limited by its linear decision boundary.
  - Decision Tree Classifier: Near-perfect accuracy of 99.99%, though likely overfitted to the data.
  - K-Nearest Neighbors (KNN): 99.94% accuracy, effective but sensitive to the choice of kkk.
  - Random Forest: Achieved 99.39%, illustrating the benefit of ensemble methods in reducing variance.
  - Support Vector Machine (SVM): Achieved 98.55%, leveraging high-dimensional data with a linear kernel.
  - Naive Bayes: 96.10% accuracy, useful in scenarios with feature independence assumptions.
  - Deep Learning: Achieved 99.09%, demonstrating strong capability to learn complex representations.
  - Extreme Learning Machine (ELM): 82.55% accuracy, limited by its simpler structure compared to deep learning.
  - Genetic Algorithm (GA): 66.33% accuracy, lower than traditional models but valuable in optimization tasks.
- Comparative Performance

Figure 7 visualizes the performance differences across models. Decision Tree, KNN, Random Forest, and Deep Learning achieved the highest accuracies, nearly 100%. GA recorded the lowest performance, likely due to the complexity of the AQI prediction task and GA's limitations for such settings.

Figure 7: Comparative Study of Model Accuracy

(Created by the authors.)



The comparative analysis shows that traditional models, particularly Decision Trees, KNN, and Random Forest, achieved the highest accuracy. The Deep Learning model also performed well and is favorable for larger datasets, given its ability to learn complex data relationships. Although GA's accuracy was lower, it remains useful in parameter optimization and specific non-convex optimization problems.

V. CONCLUSION

This study conducted a comprehensive comparative analysis of supervised machine learning, deep learning, and Extreme Learning Machine models for AQI prediction. The results demonstrated that while deep learning models, particularly LSTM networks, offer the highest predictive accuracy, their computational intensity may limit practicality in certain contexts. Supervised machine learning models, especially Random Forests, provide a dependable balance of accuracy and efficiency. ELM models present a viable option for real-time applications, delivering rapid predictions with acceptable accuracy levels.

The findings suggest that no single model is universally optimal, but rather, the choice of model depends on the specific requirements of accuracy, computational efficiency, and real-time applicability. Hybrid approaches that integrate the strengths of different models could further enhance AQI forecasting capabilities.

Some of the practical limitations include:

Data Dependency: Deep learning models require extensive historical data, which may not be available for all regions.

Computational Costs: Real-time deployment of deep learning models demands significant processing power.

Generalization Issues: Some models may struggle with new pollution patterns not present in training data.

Future Research Directions: Exploring hybrid models that combine deep learning with optimization techniques, and incorporating more real-time data streams for better model adaptability.

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